Policy Research Working Paper 11163

Inequality, Education, and Occupational Change in the Philippines

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Abstract

Despite significant progress in reducing poverty, the Philippines continues to face high inequality, which stayed elevated in the early 2000s as the economy grew. Although inequality has gradually declined since 2012, it remains among the highest in Southeast Asia. This paper examines how changes in education levels and occupational structure have shaped the wage distribution over the past two decades, particularly how changes in the relative supply of skills and the structure of employment have influenced wage gaps in recent years. Using two decades of labor force survey data, the paper examines the wage premium and the supply of skilled workers in the Philippines, finding that the slow growth in college-educated workers has sustained high wage premium for skilled workers. Unconditional quantile regressions reveal that returns to both college education and high-skill occupations increase monotonically over the wage distribution, contributing to the persistence of inequality. Changes in occupational structure have also influenced income distribution. Low- and middle-skilled jobs saw relative wage gains from 2002 to 2012, but middle-skilled occupations experienced the highest growth from 2012 to 2016—a key driver behind falling wage inequality. Employment trends followed a similar pattern, with middle-skilled job growth peaking in 2012-2016. Recent trends suggest a shift away from middle-skilled jobs, though it remains uncertain whether this reflects structural changes in the labor market or temporary disruptions.

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1. Introduction

Over the past three decades, the Philippines has made significant strides in reducing poverty, driven by sustained economic growth, structural transformation, and expansion of education. The national poverty rate declined markedly from 49.2 percent in 1985 to 16.7 percent in 2018, reflecting the synergies between economic expansion and social development. However, the COVID-19 pandemic disrupted this progress, increasing the poverty rate to 18.1 percent by 2021, underscoring the challenges of sustaining gains amidst global shocks and labor market disruptions. Poverty declined again in 2023 to 15.5 percent.

Economic growth has been an important driver of poverty reduction. Between 1985 and 2023, gross domestic product (GDP) expanded at an average annual rate of 4.1 percent, or 2.1 percent per capita, despite episodes of economic volatility. Growth was particularly robust in the years leading up to the pandemic, reflecting structural adjustments and improving macroeconomic fundamentals. The pandemic-induced recession in 2020 resulted in a 10.5 percent decline in GDP per capita, but the economy rebounded with 5 percent growth in 2021, followed by strong performance of around 7 percent in 2022 and approximately 5 percent in both 2023 and 2024, reflecting a steady and resilient recovery.

The economic structure of the Philippines, like many Asian economies, has undergone profound transformation over the past decades. Agriculture's share of employment and value-added has steadily declined, giving way to the expanding services sector and, to a lesser extent, industry. This structural shift reflects a broader transition of labor from agriculture to wage work, particularly in services, driving sustained economic growth and improving living standards. These changes were further supported by advancements in educational attainment, as the share of the workforce aged 25 and older with primary education or less fell from 55 percent in the late 1980s to 25 percent in 2024, while those with some college or higher education rose from 20 percent to about 30 percent over the same period. However, significant gaps persist, with only 20 percent of the workforce completing college by 2024, highlighting ongoing challenges in skill development and upward mobility.

Income inequality has been a persistent challenge in the Philippines, even amid significant reductions in poverty. During the 1997–1998 Asian financial crisis, inequality surged, with the income Gini coefficient reaching 47.5 percent in 1997, and remained elevated through the early 2000s despite rapid economic growth. However, since 2012, inequality has steadily declined, driven by pro-poor economic growth and increased labor transitions from agriculture to wage and nonfarm employment, particularly among households at the lower end of the income distribution. By 2023, the Gini coefficient had narrowed to 39.3 percent, down from 46.5 percent in 2012, marking a notable improvement. Despite this progress, income inequality in the Philippines remains among the highest in Southeast Asia. As of 2023, the bottom 50 percent of the population held only 23 percent of national income, while the top 10 percent captured 33 percent, with average per capita income in the top decile approximately six times higher than in the bottom, highlighting pronounced income disparities.

A decomposition of income inequality based on household attributes reveals that education gaps remain the largest contributor to its persistence, followed by occupational disparities. In 2023, differences in the educational attainment of household heads accounted for about 30 percent of total income inequality, a contribution that has remained at this level for nearly three decades

despite the progress in expanding access to education. Inequality between households based on the occupation of their head ranked as the second-largest contributor, accounting for about 20 percent of inequality in 2023, down from 32 percent in the early 2000s, reflecting some reduction in inequality across occupational groups. An analysis of inequality by sources of income further indicates that wage income has consistently been the largest contributor, accounting for nearly 50 percent of the income Gini coefficient since 2000. Trends in wage inequality closely mirrored those of overall income inequality, with the wage Gini coefficient stabilizing at 40 to 41 percent from 2002 to 2012 before declining steadily to 32 percent by 2024.

This paper examines how changes in educational attainment and occupational structure have shaped wage distribution in the Philippines over the past two decades. It focuses on how shifts in the relative supply of skills, transformations in the structure of work, and variations in returns to skills have influenced changes in wage inequality. By exploring these dynamics, the paper aims to shed light on how labor market developments and educational advancements have contributed to improvements in wage distribution and the broader reduction of income inequality in recent years.

There is an extensive body of literature examining shifts in educational attainment and occupational structure, particularly their impact on wage inequality. Evidence on these patterns is frequently characterized through the classification of occupations by skill level, with an emphasis on the role of job polarization driven by technology in influencing inequality. Job polarization was first observed and has been extensively analyzed in the United States where Autor et al. (2003) in particular documented structural shifts in the labor market characterized by the simultaneous growth of high-skilled, high-wage occupations and low-skilled, low-wage service jobs, while middle-skilled routine occupations declined. Acemoglu and Autor (2011), Autor and Dorn (2013), and Autor (2019) build on several papers[1] by constructing task-based models, which group occupations into high-skill, middle-skill routine, middle-skill nonroutine, and low-skill. They find evidence that wage inequality has increased due to job polarization, which has been driven by technological advancements automating routine tasks and enabling the offshoring of certain occupations in the United States. As a result, the college-wage premium continued to rise despite the increasing supply of college-educated workers. Similarly, Bárány and Siegel (2018) utilize these occupational groupings, but find that polarization began before the widespread adoption of information and communication technology (ICT), suggesting that earlier technological advancements such as mechanization, as well as sectoral shifts, contributed significantly to occupational wage changes in the U.S. Autor et al. (2020) add further insights by characterizing the changes in wage premia as a race between education levels and technology, where rising demand for high skilled workers competes with an increasing supply of college-educated workers. This framework explains much of the variation in education-related wage premia in the 20th century. However, in more recent years, they find that wage inequality has increasingly occurred within rather than between educational groups in the United States. Firpo et al. (2018) propose an alternative approach that uses a reweighting procedure and recentered influence function (RIF) regressions to examine the factors driving the polarization of male wages in the United States from the 1980s to the 2010s. Their findings highlight unions, occupations, and education as significant determinants of changes in wage distribution across different quantiles over this period.

[1] These include several foundational papers classifying occupations such as Autor et al. (2003) and Acemoglu and Zilibotti (2001).

Research on changes in occupational structure is particularly rich in the United States, but similar patterns are well-documented in other economies, including the United Kingdom, Germany, Norway, and Denmark (Goos et al. 2009; Spitz-Oener 2006; Michaels et al. 2014; Autor 2019). Han et al. (2023) examine South Korea's labor market between 1994 and 2008 and find that the college wage premium continued to rise despite an increasing supply of college-educated workers. This continued rise in the college wage premium was driven by technological change, which heightened demand for skilled labor, and by growing trade with China, which shifted economic activity from lower-skilled industries to more skill-intensive exports. The role of labor market demand in shaping wage premia is particularly evident in Latin America, where Acosta et al. (2019) examined 16 countries with gradually increasing education levels and found that while the supply of skilled (tertiary) and semi-skilled (secondary) workers steadily grew, their wage premia followed divergent trajectories. Returns to tertiary education rose in the 1990s, declined sharply in the 2000s, and continued to fall—albeit more gradually—through the 2010s while returns to secondary education steadily decreased through the entire period, suggesting that education premia were more influenced by employer demand than by the supply of graduates. Further underscoring the importance of labor market demand, Caner et al. (2022) documented how the rapid expansion of university slots in Turkey led to a declining wage premium among younger graduates, while the premium for older graduates remained stable, indicating that the decline in college wage premiums reflects both increased supply and potential differences in skill levels among new graduates in a less competitive environment.

Evidence on occupational structure, returns to skills and inequality in developing countries is more limited than in high-income countries. Building on Autor (2014), a large cross-country comparison of occupational patterns from 1995 to 2012 reveals increasing employment polarization in several low- and middle-income countries, including Guatemala, Panama and Turkey (World Bank 2016). In these countries, the share of workers in high-skilled and low-skilled occupations has increased, while employment in middle-skilled occupations has decreased. However, the same analysis finds a different trend emerged in countries such as Ethiopia and Nicaragua, where the growth in high-skilled employment was accompanied by an expansion in middle-skilled occupations. Gochoco-Bautista et al. (2013) used the Foster-Wolfson bi-polarization index and the Duclos, Esteban, and Ray index to measure income polarization in several Asian countries, finding no clear cross-country trend in polarization including a slight decrease in income polarization in the Philippines. However, they find that shifts in inequality are likely affected by differences in income premiums related to skills as wages grew faster for those with tertiary or higher education compared to those with lower levels of education from the mid-1990s to the mid-2000s.

With the exception of Gochoco-Bautista et al. (2013), recent literature on inequality in the Philippines has primarily examined factors such as low levels of human capital, economic changes, and remittances—either as drivers of inequality or as forces that sustain it. Tuano and Cruz (2019) explore why wealth and income inequality have persisted despite the Philippines experiencing significant economic growth and poverty reduction. They identify several contributing factors, including spatial disparities in employment growth, challenges faced by small and medium enterprises (SMEs), and political economy constraints. A key argument in their work is that the structural transformation of the Philippine economy has resulted in a "hollowed-out" industrial sector. Over the past few decades, the country has transitioned from an industry-oriented economy in the 1980s to one dominated by services, but where sectoral labor productivity has not substantially improved. While employment in services has expanded significantly, job growth in agriculture and industry has remained stagnant, limiting pathways for social mobility and

contributing to persistent inequality. Balisacan (2019) supports this conclusion, arguing that the shift toward low-productivity services has constrained opportunities for equitable growth and, by extension, wage growth for low-skilled workers. Similarly, Balisacan and Fuwa (2004) find that income variation is largely driven by differences in human capital, employment sector, and access to infrastructure. Focusing on the role of remittances—a salient factor in the Philippine context—Pernia (2008) finds that while international remittances have raised incomes across all segments of the population and significantly reduced poverty, they have also contributed to rising inequality, as wealthier households are more likely to receive remittances and in larger amounts.[2] In contrast, domestic remittances appear to be more welfare-enhancing for lower income households. World Bank (2022) corroborates these findings, providing a comprehensive analysis of inequality in the Philippines. The report examines drivers such as inequality of opportunity, regional differences, transfers and a special focus on the relationship between shifts in skills supply and employment structure with inequality. It also finds that while both international and domestic transfers have significantly reduced poverty, they have had opposing effects on inequality—with international remittances widening income gaps and domestic remittances mitigating inequality.

Despite a rich body of literature on inequality in the Philippines, the relationship between education, employment structure, and wage inequality remains underexplored. This paper addresses this gap by leveraging harmonized individual- and household-level survey data spanning two decades from 2002 to 2024, constructing a unique occupational crosswalk to track occupational trends over time and expanding the initial analytical work of Belhaj Hassine, Fernandez and Jandoc (2022). Drawing upon the frameworks of Acemoglu and Autor (2011), Autor (2019), and Bárány and Siegel (2018), and integrating Firpo et al.'s (2018) RIF procedure, this study extends existing research by analyzing education and occupational shifts, and wage premia within the context of the Philippine labor market.

The remainder of the paper is structured as follows: Section 2 describes the data and methodology used to assess structural labor market changes and their implications for income inequality. Section 3 presents the empirical findings, focusing on changes in occupational structure, skill distribution and wage disparities. Section 4 concludes with a discussion of key insights and policy implications.

2. Data and Methodology

The analysis primarily relies on data from the Labor Force Survey (LFS) covering the period from 2002 to 2024 and is complemented by information from recent rounds of the Family Income and Expenditure Survey (FIES), both produced by the Philippine Statistics Authority. The Philippine LFS, first conducted in 1956, is a quarterly nationwide survey conducted in January, April, July and October of each year, with each round typically covering at least 40,000 households.[3] Designed to collect reliable and accurate statistics on both levels and trends of labor and employment and to provide a framework for formulating labor market policies, the survey gathers employment, unemployment and underemployment data, along with data on the population's

[2] Pernia (2008) notes that remittances improve household investment in education and capital formation among poorer households. This may mitigate inequality in the long term assuming that educational attainment of children in higher income households is less influenced by remittances than in lower income households.

[3] In 2021, the PSA began collecting the monthly LFS, a shortened version of the quarterly LFS, to better assess the impacts of the COVID-19 pandemic on labor and employment.

demographic and socioeconomic characteristics. Despite its long history, the LFS only began publishing wage data from 2002 onwards. The FIES, on the other hand, is designed to be the main source of family income and expenditure data in the Philippines. Collected once every three years, before shifting to once every two years beginning in 2021, the survey is split into two interview windows and is a rider survey to the July and January rounds of the LFS. The first visit, in July, covers income and expenditure data from January to June of the same year, while the second visit, in January, covers data from July to December of the previous year. The FIES provides data for some of the country's most important statistics for development, with the data used in measurements of poverty and the Human Development Index (HDI), as benchmark information for weights used in the estimation of the Consumer Price Index (CPI) as well as in estimating the country's poverty incidence and threshold (PSA 2021).

The Philippines uses the Philippine Standard Occupational Classification (PSOC) to record the occupational categories of its workers. Since 2001, the LFS has used two versions of PSOC to adapt to the changing nature of work and to align with international standards: the 1992 PSOC, used in LFS surveys from January 2001 to January 2016, was patterned after the International Standard Classification of Occupations 1988 (ISCO-88) while the 2012 PSOC, adopted from April 2016 and still currently in use, is based on ISCO 2008 (ISCO-08). As the period of analysis in this study encompasses both PSOC versions, the occupation codes needed to be harmonized to create continuity across rounds. An occupation crosswalk was constructed, harmonizing the two-digit occupation codes used across LFS rounds to create 22 unique occupation codes that are as consistent as possible across time given available data. The occupation codes were then grouped into broader occupation categories, where High-skill occupations comprise managers and managing proprietors, professionals, and associate professionals and technicians; Middle-skill routine occupations include clerical support workers, craft and related trades workers, sales workers, and plant and machine operators and assemblers; Middle-skill nonroutine occupations include service workers, while Low-skill occupations encompass elementary occupations, such as cleaners and helpers; and laborers in mining, construction, manufacturing and transport. While the process of creating balanced occupation codes broadly follow the occupational classification of Dorn (2009), Acemoglu and Autor (2011) and Bárány and Siegel (2018), the harmonization used in this paper was adjusted to better suit the country context.

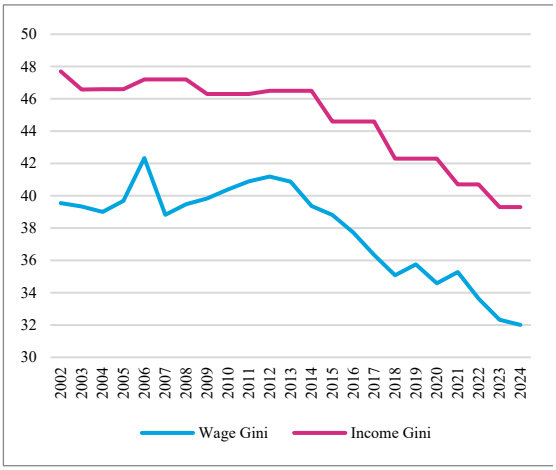
We focus the analysis on a sample of employed wage workers aged 15 and above. As such, the analysis excludes the following categories of workers which the PSA classifies as non-wage workers: 1) self-employed without any paid employee, 2) employer in own family-operated farm or business, and 3) worked with or without pay in own family-operated farm or business.[4] Following Bárány and Siegel (2018), workers with missing wage data and those in armed forces occupations were excluded from the analysis.

By focusing on employed wage workers, the analysis identifies key trends in income and wage inequality and examines the roles of labor supply, educational attainment, skill valuation, skill availability and occupational composition in shaping the earnings distribution over time. Broadly, income and wage inequality in the Philippines have followed similar trends over the past two decades, which can be divided into two distinct phases.

[4] As mentioned, the LFS only began publishing wage data beginning in 2002, which limits the analysis to cover only the two decades from 2002-2024.

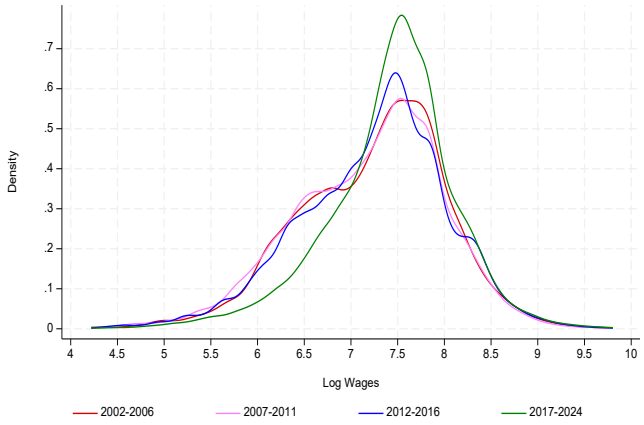
The first phase (2002–2012) was characterized by rising inequality, with both wage and income Gini coefficients increasing. This period followed the aftermath of the 1997–1998 Asian financial crisis, during which inequality surged and remained elevated despite rapid economic growth. The second phase, beginning in 2012, marked a reversal of this trend, with inequality entering a steady decline that accelerated in recent years (Figure 1). The density of log wages in Figure 2 further supports this shift, showing a more compressed wage distribution in the later years, indicating reduced wage dispersion.

Figure 1. Income and Wage Inequality trends 2002-2024



Source: Labor Force Survey (LFS) 2002–24 and FIES 2000-2023.[5]

Figure 2. Density of Log Wages 2002-2024



Source: LFS 2002–2024.

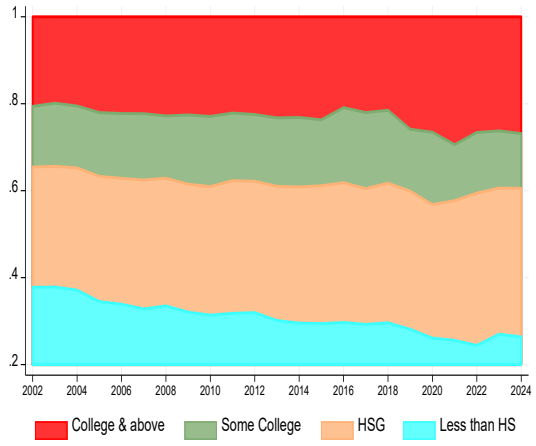
One way to understand the patterns of earnings distribution over time is to begin by analyzing changes in labor supply, educational attainment, and wage earnings, and then delve deeper into the valuation of skills, skills supply, and occupational composition. The Philippine labor market has undergone significant transformations over the past two decades, characterized by an expansion of wage employment and rising educational attainment. Between 2002 and 2024, the share of workers engaged in wage employment increased from 48 percent to 68 percent, with growth observed among both men and women. Male wage employment rose from 49 percent to 70 percent, while female wage employment increased from 47 percent to 64 percent. However, overall employment rates remain lower for women, at 47 percent in 2024 compared to 70 percent for men. The composition of the labor force has also shifted toward higher educational attainment. The share of hours supplied in wage employment by workers with less than a high school education declined from 38 percent to 26 percent, while the share of hours supplied by workers with a college degree or higher increased from 21 percent to 27 percent (Figure 3).[6] The rise in educational attainment has been particularly pronounced among women. By 2024, 40 percent of female hours in wage employment were supplied by college-educated workers, compared to 19 percent of male hours—up from 31 percent and 14 percent, respectively, in 2002 (Figure 1A in the appendix).

[5] Since income Gini data from FIES is available only every three years (and every two years since 2021), we assumed constant values between survey years.

[6] The hours shares are estimated following Autor (2019) approach and reflect an equilibrium outcome shaped by labor market dynamics, not latent labor supply.

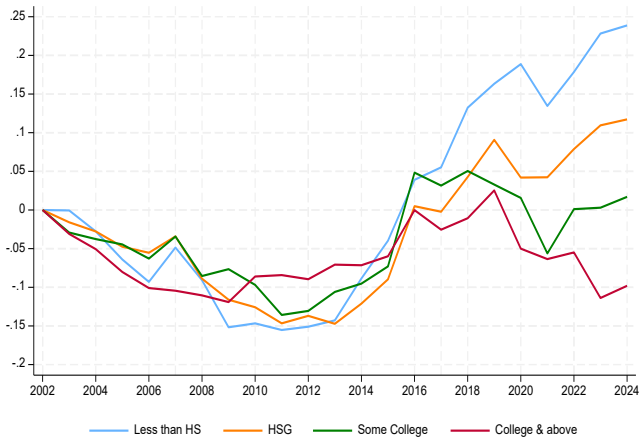
Real wage trends have evolved unevenly across education groups over the past two decades. Between 2002 and the early 2010s, real wages declined for all workers, remaining below their 2002 levels until 2016 (Figure 4). The decline was more pronounced among non-college-educated workers, whose real wages fell by around 15 percent between 2002 and 2012, compared to a 10 percent decline for those with some college or higher education. From 2012 onward, real wages began to recover, with non-college-educated workers experiencing faster growth—about 32 percent between 2012 and 2024—compared to just 5 percent for college-educated workers, contributing to narrowing wage inequality. The fastest wage increase for non-college-educated workers occurred between 2012 and 2016 (16 percent), partly driven by transitions from low-skill to medium-skill occupations. Since 2016, wage growth has decelerated, marked by greater volatility and employment fluctuations across skill levels. Despite these shifts, substantial wage disparities persist. By 2024, the real wages of non-college-educated workers were 18 percent higher than in 2002, while earnings for workers with some college or higher education saw no improvement. Nonetheless, the average real wage of workers with some college or higher remained nearly twice as high as that of non-college-educated workers in 2024.

Figure 3. Share of Hours Worked by Education Group



Source: LFS 2002–2024.

Figure 4. Composition-Adjusted Real Log Weekly Wages



Source: LFS 2002–2024.

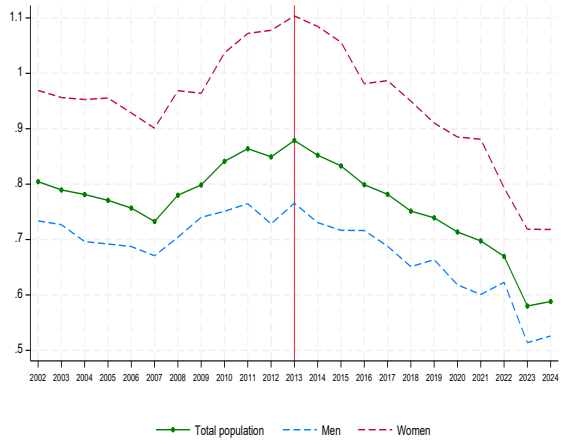
The observed trends in labor supply and wage earnings highlight broader structural shifts in the labor market, which need to be analyzed in conjunction with the evolving valuation of skills, as reflected in the college wage premium, the relative supply of skilled labor, and occupational changes. The next section will explore these dynamics, focusing on the factors contributing to the persistence of a high college wage premium, its relationship to changes in the relative supply of skills, and how shifts in occupational composition are influencing wage trends across different education or skill groups. This analysis will provide deeper insights into the forces shaping earnings inequality and labor market outcomes.

3. Skill Premium, Skill Supply, and Occupational Shifts

A useful starting point for understanding the pattern of wage inequality is to analyze trends in the college wage premium—the wage gap between college graduates and high school graduates,

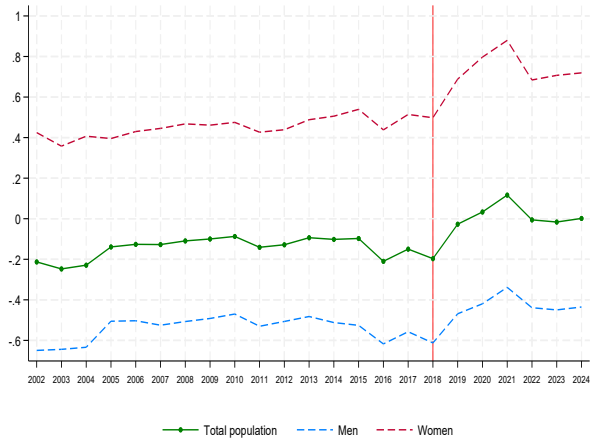
which can serve as a proxy for the market's valuation of skills.[7] As shown in Figure 5, the college wage premium has remained persistently high, though it has fluctuated over time. It declined during 2002–2007, rose sharply to a peak of 88 percentage points in 2013, and then steadily fell to 59 points by 2024. In that year, the average college graduate earned 80 percent more than the average high school graduate, while workers with some college education earned 53 percent more. Although the college premium has followed similar trends for both men and women, it has been consistently higher for women, indicating that disparities in returns to skills may be a more significant driver of inequality among women than among men. The trends in the college wage premium closely mirror broader patterns in wage inequality, suggesting that it plays an important role in sustaining inequality over time. The college wage premium is influenced by several factors, including the relative supply of skilled workers. Figure 6 suggests that the slow growth in the supply of college-educated workers, coupled with a persistent skills shortage, has maintained the scarcity value of skills. This has kept the skill premium high and hindered faster progress in reducing wage inequality.

Figure 5. Composition-Adjusted College/ High-School Log Wage Ratio, 2002–24



Source: LFS 2002–2024.

Figure 6. College/High-School Log Relative Supply, 2002–24



Source: LFS 2002–2024.

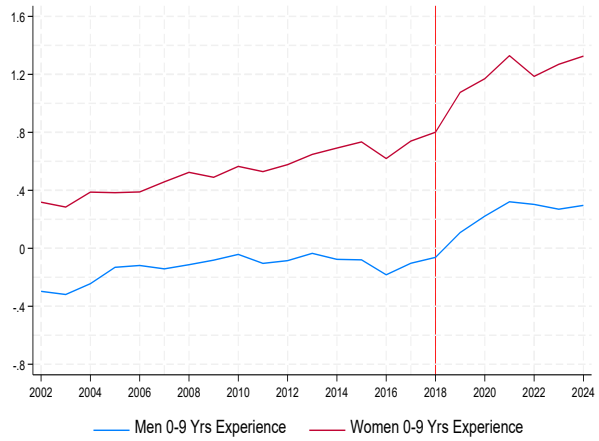
The supply of college-educated workers has seen only modest growth in recent years. Between 2002 and 2010, it increased marginally, followed by a slow decline from 2010 to 2018. However, it picked up again during 2018–2021 before declining slightly in 2024. The supply of college-educated workers has been consistently higher for women than for men, and it accelerated more rapidly for women during 2018–2021. This recent increase was driven primarily by young college graduates entering the labor market, with a particularly sharp rise in the relative supply of young women with college degrees (Figure 7.A). Figure 7.B further shows that the supply of experienced college graduates—those with 20–29 years of potential experience—grew only marginally, with most of the growth occurring among women. The gradual rise in tertiary education enrollment, especially among women, has contributed to a noticeable increase in the average education level of the labor force in recent years. However, despite these gains, the economy continues to face a

[7] We estimate the college premium following Acemoglu and Autor (2011) by regressing log weekly wages for full-time workers annually on education dummies, experience, and gender. Composition-adjusted mean log wages are calculated at specific experience and education levels, weighted by average employment shares. The yearly ratio of mean log wages for college graduates to high school graduates is plotted over time.

significant skills shortage. This shortage not only constrains economic growth and productivity but also sustains a high skill premium, reinforcing persistent wage inequality.

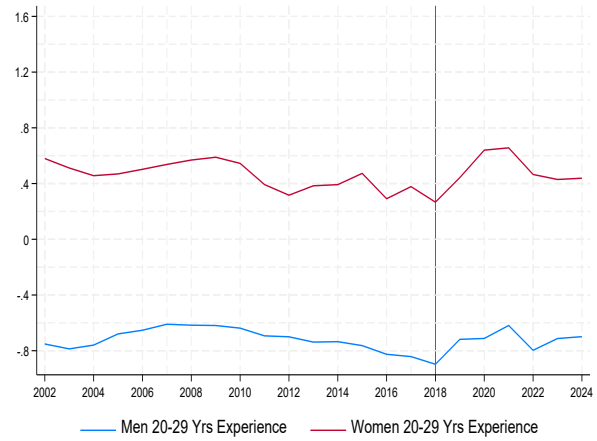
Figure 7. College/High-School Log Relative Supply by Age, 2002–24

A. Young Cohort



Source: LFS 2002–2024.

B. Older Cohort



Source: LFS 2002–2024.

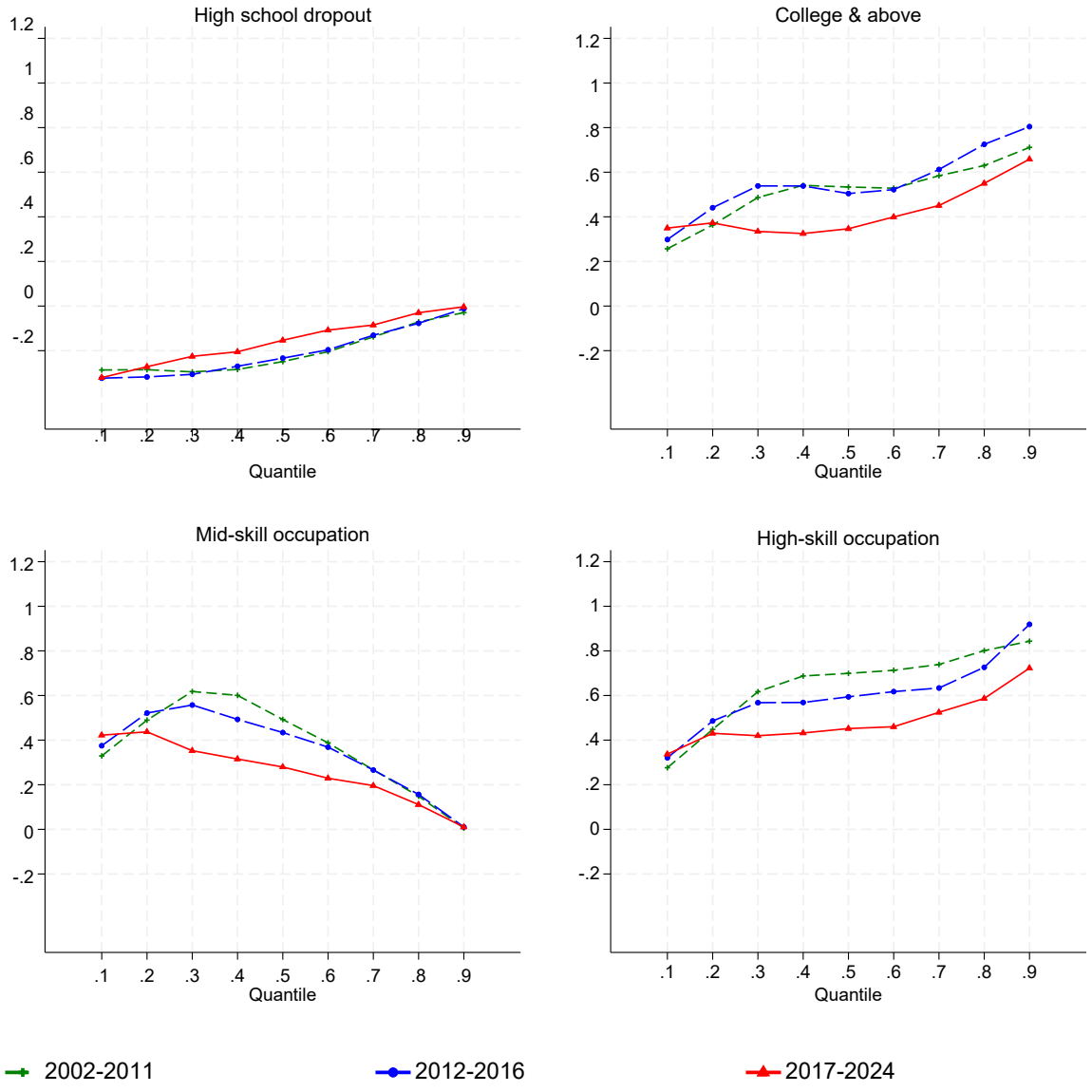
The college wage premium and skill supply may not fully capture important dimensions of inequality. Expanding the analysis to examine returns across the income distribution can offer a better understanding of how education and occupations contribute to inequality. These returns are estimated using Recentered Influence Function (RIF) regressions of unconditional quantiles of log real weekly wage income (Firpo et al. 2018), to examine how the rewards of education and skills vary across income groups. The analysis shows that increasing returns to college education and high-skill occupations across the income distribution play an important role in perpetuating wage inequality. As shown in Figure 8, returns to both college education and high-skill occupations rise monotonically with income percentiles, a pattern that has persisted over the past two decades, though with slight moderation in recent years (see also Figure 3A in the appendix). This indicates that higher-income individuals benefit disproportionately from higher education and high-skill jobs, further widening the wage gap. While the persistence of high returns to college education reflects the labor market's valuation of skills, the disparity in returns between high- and low-income groups is concerning. One explanation for this disparity is differences in school quality: individuals from lower-quality educational backgrounds tend to achieve lower labor market returns for the same level of education (Patrinos et al. 2006), exacerbating inequality.[8]

As shown by the RIF-regression coefficients for the 10th, 50th, and 90th quantiles and the wage income Gini coefficient in Tables 1A and 2A in the appendix, the inequality-enhancing effect of college education and high-skill employment is particularly pronounced at the top of the income distribution, where higher returns are concentrated. The inequality-enhancing effect of college education appears to be increasing over time, as evidenced by its positive impact on the wage Gini

[8] A more general explanation lies in the complementary relationship between higher education and unobserved endowments, such as innate abilities, access to quality schooling, or social networks, which are more prevalent among higher-income groups (Buchinsky 1994; Mwabu and Schultz 1996). As a result, college education not only reinforces existing disparities but also amplifies them, as it complements rather than compensates for these unobservable advantages. The analysis of returns to education and occupations across household income groups reveals a more pronounced inequality-enhancing effect of college education and high-skill occupations (see World Bank 2022).

coefficient in Table 1A, while the impact of high-skill occupations remains relatively unchanged. In contrast, secondary education and mid-skill occupations generally reduce inequality. While mid-skill jobs contribute to narrowing the wage gap, particularly at the higher end of the income distribution, they have a modest inequality-increasing effect at the lower end.

Figure 8. Returns to Education and Occupation by Quantile



Source: LFS 2002–2024.

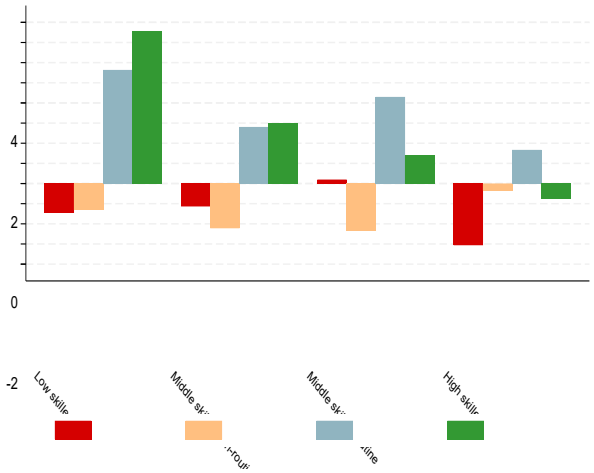
The 2010s were marked by a relative expansion of middle-skilled occupations and an increase in wages for low- and middle-earning groups, contributing to a moderation in wage inequality. Following Dorn (2009), Acemoglu and Autor (2011) and Bárány and Siegel (2018), we classify occupations into four broad categories—high-skill, middle-skill routine, middle-skill nonroutine, and low-skill— to examine changes in employment and wage trends. Between 2002 and 2012, real wages declined across all occupation groups, with the steepest drops occurring in middle- and

high-skilled occupations (Figure 9). However, from 2012 onward, wages began to recover, with the most significant increases occurring for low- and middle-skilled workers. The period from 2012 to 2016 saw the highest wage growth for middle-skilled occupations, contributing to a decline in wage inequality. Employment trends mirrored this pattern, as middle-skilled occupations expanded significantly while the shares of low- and high-skilled employment declined (Figure 10).

Beginning in 2002, employment in the Philippines was concentrated in middle- and low-skilled occupations, with 55 percent of total hours in middle-skilled jobs (mainly routine) and 27 percent in low-skilled jobs. Over the following two decades, high-skilled employment fell by about 1 percentage points, from 19 to 18 percent of hours, while middle-skilled employment grew, peaking between 2012 and 2016 with a 3.6 percentage point increase. This shift was accompanied by a decline in low-skilled employment, which fell by 2.3 percentage points, from 26.8 to 24.5 percent, reflecting a movement of workers into middle-skilled occupations. However, from 2016 to 2024, middle-skilled employment declined by 2.8 percentage points, driven primarily by a reduction in nonroutine middle-skill jobs. This decline was offset by rising employment in low-skilled occupations and, to a lesser extent, high-skilled occupations.

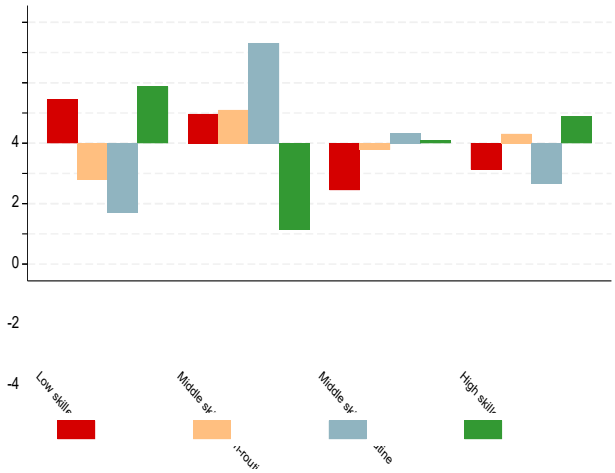
While the early 2010s suggested an upward shift, with more workers moving into middle-skilled jobs, recent trends indicate a movement of the middle-class toward lower-skilled employment, evidenced by the disproportionate growth in total hours worked in low-skilled occupations and a decline in middle-skilled occupations. This pattern raises concerns about the potential hollowing out of middle-tier jobs and the risk of economic polarization if the trend persists. While the decline in middle-skilled employment could reflect long-term technological shifts, as observed in advanced economies (Autor, 2019), it is unclear whether this trend is structural or driven by short-term cyclical factors, including the impact of COVID-19. This uncertainty underscores the need to monitor whether the earlier gains in reducing inequality were temporary and whether technological progress and digitalization are contributing to a longer-term polarization of the labor market. Nonetheless, the period since 2012—marked by relative wage gains for low- and middle-earning groups—has played an important role in reducing wage inequality, highlighting the importance of labor market dynamics in shaping income distribution.

Figure 9. Change in Log Wages by Occupation, 2002-2024, Percent



Source: LFS 2002–2024.

Figure 10. Changes in Occupational Employment Shares, 2002-2024, Percentage Points

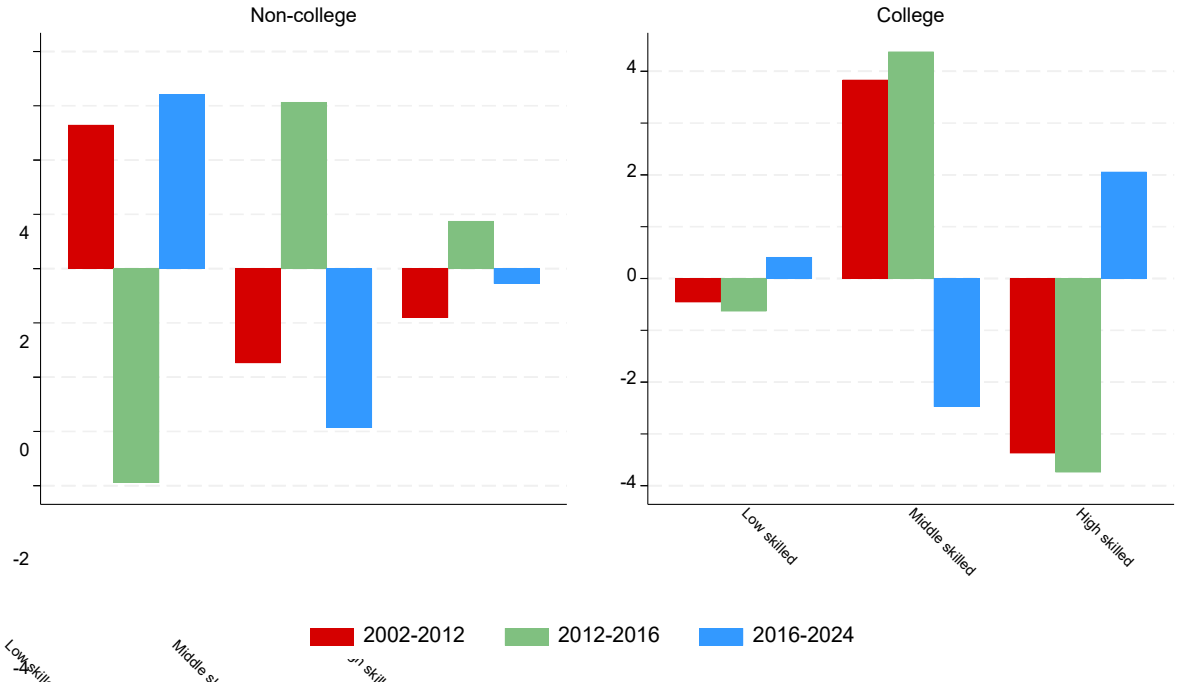


Source: LFS 2002–2024.

The analysis of occupational changes across education groups in Figure 11 provides a clearer picture of employment shifts across skill levels. Among college workers (those with some college or higher education), occupational movement between 2002 and 2024 was modestly, though not uniformly, directed toward middle-skill occupations. Over this period, the share of hours worked by college workers in high-skill occupations declined by over 5 percentage points (from 43 percent to 37 percent), while the share in middle-skill occupations increased from 49 percent to 55 percent. The share in low-skill occupations remained relatively stable at less than 8 percent. However, these shifts were not uniform over time. Most of the transitions from high-skill to middle-skill occupations occurred between 2002 and 2016, with an acceleration in 2012–2016. Since 2016, signs of polarization have emerged, with college workers reallocating from middle-skill occupations to high-skill and to a lesser extent low-skill jobs. Compared to 2016, college workers have seen a slight upward shift in their occupational distribution. By 2024, the share of hours worked in high-skill jobs was about 2 percentage points higher, while the share in middle-skill jobs was 2.5 percentage points lower.

Among non-college workers (those with high school or lower education), occupational movement between 2002 and 2024 trended downward, though modestly and non-uniformly. Over this period, the share of non-college workers in high-skill occupations declined from 3.2 percent to 2.9 percent, while the share in middle-skill occupations fell from 58 percent to 57 percent. Conversely, the share in low-skill occupations increased by almost 2 percentage points (from 38.5 percent to 40.4 percent). Between 2002 and 2012, there was a notable shift of non-college workers from middle-skill to low-skill occupations. However, this trend reversed during 2012–2016, as the share of non-college employment in low-skill occupations fell by almost 4 percentage points, offset by a 3 percentage point increase in middle-skill occupations (from 57 percent to 60 percent), while the share in high-skill occupations saw a slight rise (from 2.3 percent to 3.2 percent). From 2016 to 2024, the trends shifted again, with middle-skill employment declining by about 3 percentage points, and low-skill employment increasing by the same proportion.

Figure 11. Changes in Occupational Employment Shares by Education Groups, 2002-2024, Percentage Points



Source: LFS 2002–2024.

Recent trends suggest a potential shift toward occupational polarization, with non-college workers increasingly moving from middle-skill to low-skill jobs, while college workers are modestly transitioning from middle-skill to high-skill jobs. These changes may be linked to ongoing technological and digital advancements, which tend to complement the skills of more educated workers. While technology tends to transform the nature of work for all, its effects can be more pronounced for less-educated workers, who experience a decline in opportunities for specialized, higher-paying middle-skill jobs. The pandemic may have accelerated these trends, with increased digitization, though the full extent of these changes remains to be seen as they require a longer time span for robust analysis.

The shift of non-college-educated workers toward middle-skill and, to a lesser extent, high-skill occupations between 2012 and 2016 likely contributed to faster real wage growth and a reduction in wage inequality. However, from 2016 to 2024, these gains partially reversed, as middle-skill employment declined and less-educated workers increasingly moved into low-skill jobs, signaling early signs of occupational polarization. Despite this shift, real wages for low-skill and less-educated workers continued to rise, potentially supported by incremental minimum wage increases, helping maintain the downward trend in wage inequality.[9] While these trends have not yet negatively impacted wage inequality, the persistence of polarization could pose longer-term risks, potentially widening wage disparities and reversing recent progress in reducing inequality.

We further analyze the evolution of wages among college and non-college workers in high-, medium-, and low-skill occupations using the kernel density reweighting technique developed by DiNardo, Fortin, and Lemieux (1996), referred to as DFL. This approach helps address the

[9] This period, specifically from 2012 to 2016, seems to coincide with a sustained increase in real minimum wage rates. This topic could be explored further, but is beyond the scope of this paper.

following question: How would wages for college and non-college workers have changed between 2002 and 2024 if occupational composition had evolved as observed, but wage levels within each occupation had remained fixed at their 2002 levels?[10].

Following DFL,[11] we consider the observed wage distribution $f(w)$ in year $t0$ as the joint distribution of wages w and covariates x (such as education, occupation, gender, etc.), integrated over the domain of covariates Ωx in year $t0$:

$$f_{t0}(w) = \int_{\chi \in \Omega \chi} dF(w, x \mid t_{w,x} = t0)$$

Using iterated expectations, this expression can be reformulated as:

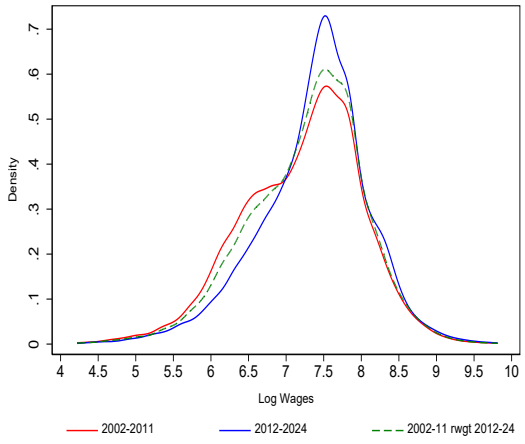
$$f_{w_{t0}}^{x_{t0}}(w) = \int f(w \mid x, tw = t0) dF(x \mid tx = t0)$$

$$f_{w_{t0}}^{x_{t1}}(w) = \int f(w \mid x, tw = t0) dF(x \mid tx = t1)$$
$$= \int f(w \mid x, tw = t0) * \psi_x(x) dF(x \mid tz = t0)$$

The function $\widehat{\psi_x} = [\Pr(tx = t1 \mid x) / \Pr(tx = t0 \mid x)] * [\Pr(tx = t0) / \Pr(tx = t1)]$, which reweights the distribution of covariates in period $t0$ to match those in $t1$ is estimated using with a logit model.

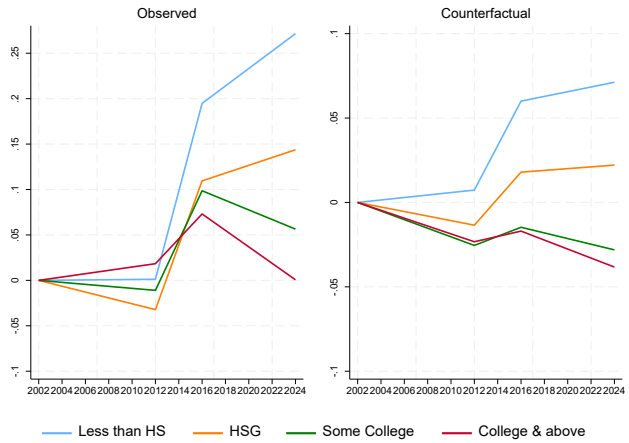
Figure 12. Observed and Counterfactual Changes in Log Wages, 2002-2024

A. Observed and reweighted wage densities



Source: LFS 2002–2024.

B. Wage Changes by education group



Source: LFS 2002–2024.

Figure 12 presents the results of the reweighting analysis, which examines how changes in occupations and workforce characteristics have influenced wage distributions over time. Panel A compares the actual wage densities for 2002–2011 and 2012–2024 with a reweighted wage distribution (dashed green line). This counterfactual distribution simulates how the 2002–2011 wage distribution might have looked if the occupational structure had mirrored that of 2012–2024. The counterfactual wage distribution is more compressed than the actual 2002–2011 distribution,

[10] To simplify the analysis and isolate the impact of changes in occupational composition on wage trends, we hold wages fixed at their 2002-2011 levels. This approach allows us to focus on the significant occupational shifts and wage changes that occurred during the 2012-2024 period.
[11] See also Autor (2019).

indicating that shifts in occupations and worker characteristics have contributed to reducing wage inequality. This suggests that occupational reallocation, particularly the movement of non-college workers into middle-skill jobs, have contributed to a more balanced wage structure over time.

Panel B offers further insights into both the observed and counterfactual changes in wages by educational attainment over the past two decades. The left figure reports the observed pattern of wage earnings, revealing distinct trends across educational groups. For college-educated workers, real wages experienced a modest increase between 2002 and 2012, followed by a more pronounced rise from 2012 to 2016. However, this upward trend reversed after 2016, with real wages progressively declining through 2024. In contrast, workers with high school or lower education experienced a decline in real wages during the 2002-2012 period, but their wages rose steeply from 2012 to 2016 and continued to increase, albeit more modestly, in the subsequent years up to 2024. These patterns suggest a narrowing of wage inequality in recent years, driven by stronger wage growth for less-educated workers compared to college-educated workers.

The right graph (Counterfactual) illustrates the effects of reweighting the 2002 wage distribution to reflect changes in occupational structure during the periods 2012–2016 and 2017–2024. Using the four broad occupation categories—low-, middle-, and high-skill occupations—for each of the four education groups as above, the analysis simulates how wage distributions would have evolved if occupational composition had shifted as observed, while holding wage levels within occupations fixed at their 2002–2011 levels. This approach isolates the impact of occupational reallocation on wage trends, providing insights into how changes in job composition have influenced wage outcomes across different education groups. It reveals that occupational reallocation account for a significant share of the rise in non-college wages after 2012, particularly during the 2012–2016 period. When examined by gender, the findings suggest that occupational changes were more influential for male non-college workers, as shifts in job structure disproportionately affected men (Figure 4A in the appendix). However, occupational reallocation does not seem to explain much of the wage trends for college-educated workers, as their wage changes are more likely driven by productivity variations, supply and demand dynamics, or other factors influencing wage levels within and between occupations.[12] Furthermore, the effect of occupational reweighting on college wages weakens after 2016, as occupational shifts among college workers were modest compared to those among non-college workers during this period.

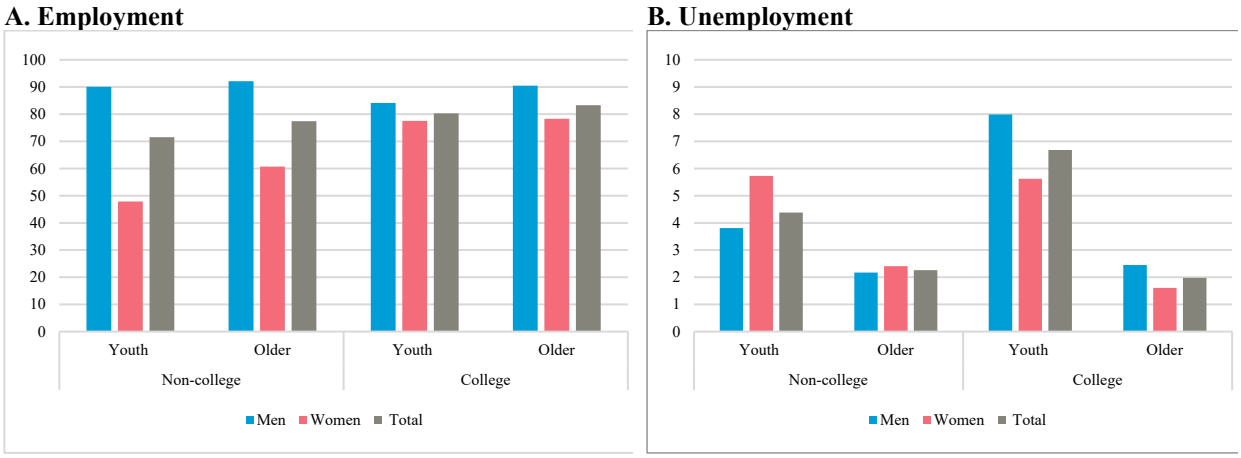
Despite large college wage premiums, completing tertiary education does not automatically lead to better labor market outcomes, particularly for young graduates. In 2024, the unemployment rate among youth college graduates aged 25 to 34 stood at 6.7 percent, significantly higher than the 4.4 percent recorded for their peers with lower levels of education. While the gap is narrower among older graduates, they too experience higher unemployment rates relative to their peers particularly among men.[13] The elevated unemployment rates among younger entrants into the workforce could be due to higher reservation wages, which can prolong job searches and contribute to unemployment and absence from the labor market. Notably, 35 percent of college graduates that

[12] The DFL decomposition approach isolates the impact of occupational shifts by holding wage levels within occupations fixed, abstracting from broader economic forces that influence wage evolution over time. See also Autor (2019).

[13] The gender gap between male and female unemployment rate is likely due to social norms around employment in which men of working ages are almost all in labor force while women's participation is more dependent on household and individual characteristics including educational attainment. See Belhaj Hassine and Fernandez. (2021) for more on female labor force participation.

were out of the labor force (excluding family reasons), were awaiting results of a job application compared to 9 percent of non-college graduates. The high rate of unemployment among young graduates, which would likely be exacerbated by an increase in tertiary graduates, suggests that expanding tertiary education alone is not a panacea to reducing inequality.

Figure 13. Employment and Unemployment by Age, Education and Gender, 2024, Percent



Source: LFS 2024.

Note: Youth are aged 25 to 34 and older are aged 35 to 54; Non-college educated are workers with high school degree or less and college educated are those who completed tertiary education.

Youth employment prospects also vary significantly by field of study, reflecting broader inequalities faced even by college-educated individuals—particularly as some fields offer more employment opportunities than others, shaping future prospects and reinforcing disparities. Graduates in disciplines such as natural sciences and mathematics appear to secure employment more easily. In contrast, some fields appear to place greater emphasis on experience or tenure. For instance, while young graduates with degrees in education experience higher unemployment rates compared to their peers, older graduates in the same field are almost universally employed.[14] However, in several of the most popular fields, unemployment remains persistently high. This is particularly evident in Information & Communication Technologies (ICT) and Engineering, Manufacturing & Construction—fields often perceived as highly valued and central to future economic growth. These trends suggest a skills mismatch, where students pursue fields they perceive as in demand, yet labor market conditions do not align with these expectations. Given this, a hypothetic policy increasing in the number of college graduates would need to account for the difference in labor market demand in varying fields of study.

[14] Conditional on remaining in the labor force.

Conclusion

The Philippines has made considerable progress in reducing poverty over the past three decades, driven by sustained economic growth, structural transformation and rising educational attainment. Like other Asian economies, the country experienced significant shifts in economic structure during this period, with agriculture contracting in terms of both employment share and value-added to the economy while the share of services and to a lesser extent, industry, expanded. These structural shifts were further supported by improvements in educational attainment, as the share of the workforce with at most a primary education fell from 55 percent to 25 percent from the late 1980s to 2024.

Despite this progress, inequality has been a persistent challenge in the Philippines. While income inequality has steadily declined since 2012, it remains among the highest in Southeast Asia with the average per capita income on the top decile approximately six times higher than the bottom decile as of 2023.

This paper seeks to examine how changes in educational attainment and occupational structure have shaped wage distribution in the Philippines over the past two decades, focusing on how shifts in the relative supply of skills, changes in the structure of work and variations in returns to skills have influenced the reduction in wage inequality observed in more recent years. By analyzing these dynamics, the paper aims to identify emerging labor market trends that could influence income inequality in the near future

The analysis draws on data from the Philippine Labor Force Survey (LFS) and the Family Income Expenditure Survey (FIES). As the study encompasses two rounds of occupational classifications, an occupation crosswalk was created to create continuity across rounds; occupations were then grouped according to broad skill classifications, following Dorn (2009), Acemoglu and Autor (2011) and Bárány and Siegel (2018), to allow for a more nuanced analysis of wage and employment trends across occupation groups.

The analysis begins with an examination of the wage premium and its relationship to the relative supply of skilled workers, focusing in particular on trends in the supply of college-educated workers over the past two decades, starting in 2002. The results reveal that while the college wage premium—the wage gap between college graduates and high school graduates—has fluctuated over time, it has remained persistently high, with the average college graduate earning 80 percent more than the average high school graduate in 2024. The persistence of the college wage premium appears closely linked to the relative supply of college-educated workers, which has only shown modest growth in recent years. Together, these results suggest that the slow growth in the supply of highly educated workers has kept the skills premium, and thus wage inequality, high.

While straightforward, this initial examination may not fully capture important dimensions of inequality. Using RIF regressions of unconditional quantiles of log real weekly wage income, the analysis is expanded to examine returns across the income distribution to provide a more nuanced understanding of how education and occupations contribute to inequality, and to examine how the rewards of education and skills vary across income groups. The results of these regressions reveal that over the past two decades, returns to both college education and high-skill occupations rose monotonically over the wage distribution, indicating that those better-off benefit disproportionately from both higher education and high-skill occupations. While the persistence of high returns to college education reflect the labor market's valuation of skills, the disparity in

returns across income groups could be concerning, particularly as it could exacerbate existing inequalities.

Shifts in occupational structure have also played a significant role in shaping the income distribution. Occupations are classified into four broad categories, encompassing high-skill, medium-skill routine, medium-skilled nonroutine and low-skilled occupations. Overall, the wage trend from 2002 to 2012 reveals that middle and low-skilled occupations experienced relative wage gains, with the period from 2012 to 2016 marking the highest wage growth for middle-skilled occupations, which contributed significantly to the decline in wage inequality. Employment trends appear to mirror this broad pattern, with the overall employment trend showing that middle-skilled occupations grew in the two decades since 2002, with its peak also occurring in the 2012 to 2016 period. However, a different trend seems to be emerging, with a shift away from middle-skilled occupations towards both high- and low-skilled occupations beginning in 2016.

An analysis of occupational changes across education groups provide a clearer picture of employment shifts across skill levels. Among higher-educated workers (those with at least some college education), the overall trend appears to be directed towards middle-skilled occupations, with the share of hours worked in middle-skilled occupations rising to reach 55 percent in 2024. Since 2016, however, there appears to be some movement away from middle-skilled occupations towards both high- and low-skilled work. In contrast, occupational movement among non-college workers (those with at most a high school education) has trended downward, with the share in low-skilled occupations rising in recent years. Notably, non-college workers also experienced an increased share in middle-skilled employment from 2012 to 2016, but this trend reversed in 2016, with the fall in both high- and medium-skilled employment accompanied by an increased share in low-skilled occupations.

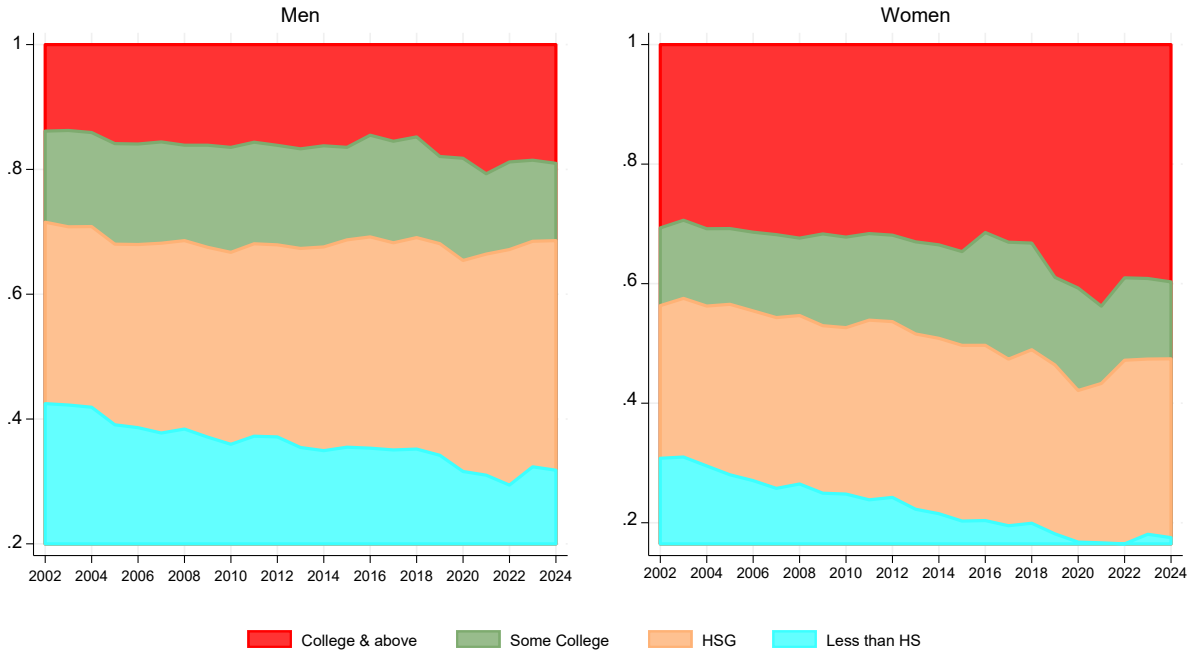
Taken together, recent trends appear to suggest the emergence of a decline in middle-skilled employment and a trend towards rising employment shares in high-skilled occupations (for college workers) and low-skilled occupations (for non-college workers). This divergence could potentially be linked to ongoing technological and digital advancements which tend to disproportionately benefit more educated workers. While technology changes the nature of work for most workers, effects can be more pronounced for less educated workers who can experience a considerable decline in opportunities for higher paying middle-skilled jobs. This could be a concern, particularly as the DFL analysis reveals that movement of non-college workers into middle-skilled occupations account for a significant share of the wage increase among less educated workers, which has contributed to a more balanced wage structure over time. However, it is currently unclear whether these emerging trends are structural or driven by short-term cyclical factors, including the impact of the COVID-19 pandemic. While the pandemic may have accelerated the trend towards digitalization, the full extent of these changes remain to be seen as they require a longer time span for robust analysis.

Policies may help mitigate significant disparities in income inequality particularly with the changing nature of work. Enhancing foundational skills, such as math and literacy, at the level of basic education can help ensure that future workers are able to adapt to changing labor market demands. Closing the quality gap in higher education could also mitigate the trends found in this paper, which show that higher-income individuals benefit disproportionately from higher education. Expanding skills development programs, in close consultation with the private sector, could help workers transition to jobs that match the needs of the labor market. Finally, expanding access to tertiary education could raise the productivity of the labor force, promote innovation and

potentially reduce wage inequality, however it is important to note that it is not a cure-all. For one, despite large college wage premiums, young college graduates have rates of unemployment that are considerably higher than peers with lower levels of education. Beyond this, employment prospects among the youth also appear to vary significantly depending on their field of study. Together, these trends suggest potential skills mismatches in the labor market. While beyond the scope of this paper, this topic could be worth exploring for future research. Subsequent studies could also explore how school quality influences labor market outcomes as well as trends in the potential polarization of jobs in the Philippines.

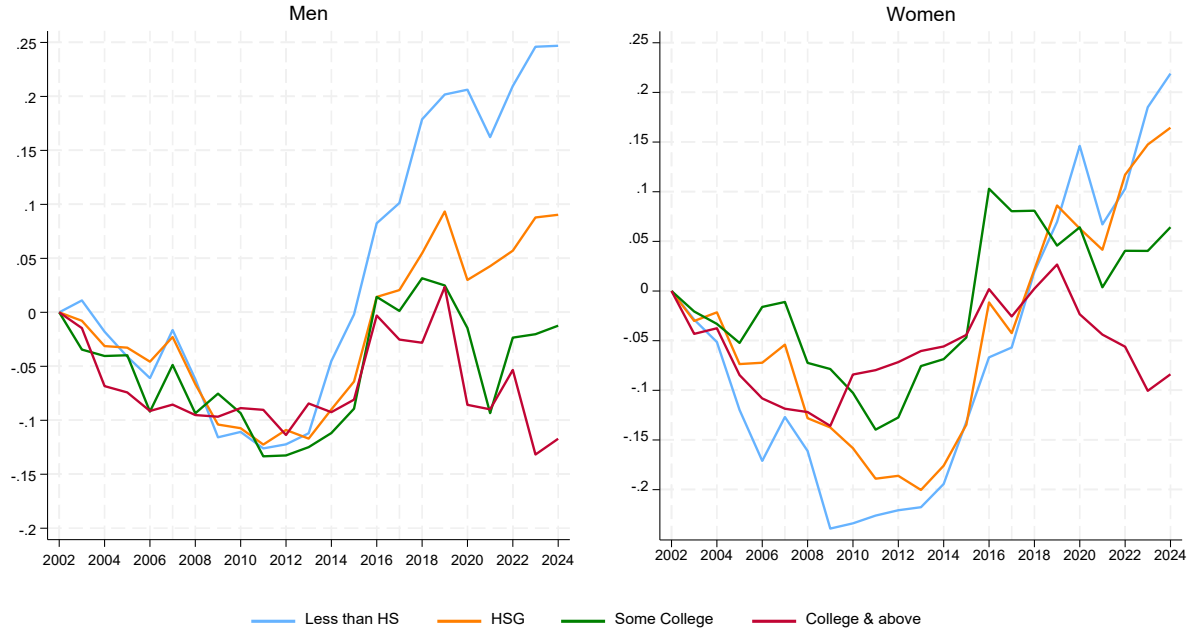
Appendix

Figure 1A. Share of Hours Worked by Education Group and Gender, 2002-2024



Source: LFS 2002–2024.
Note: The sample consists of all wage-employed individuals aged 15 and older, who work in nonfarm and nonmilitary sectors.

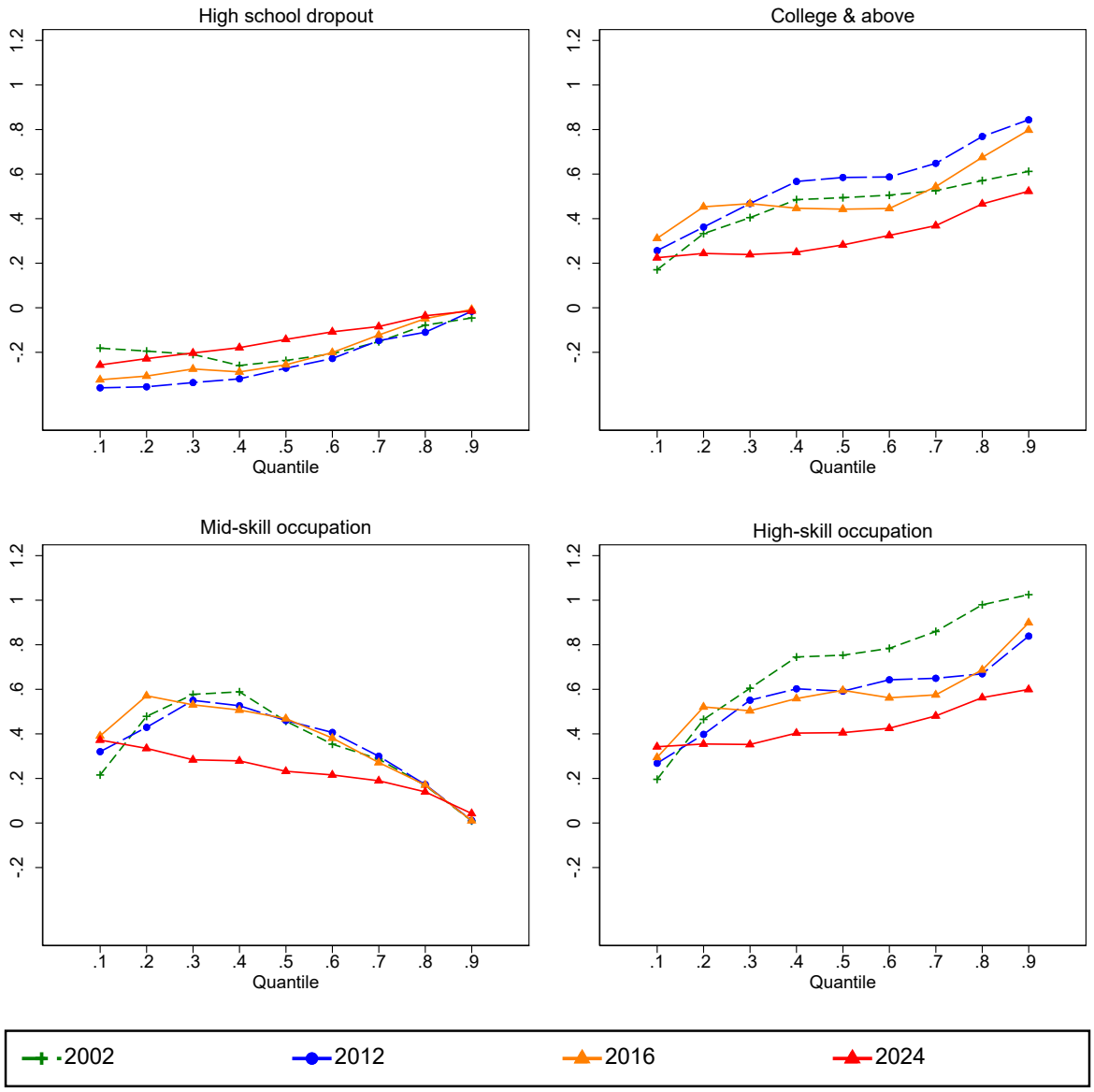
Figure 2A. Composition-Adjusted Real Log Weekly Wages by Education Group and Gender, 2002-2024



Source: LFS 2002–24.

Note: The figure illustrates cumulative changes in real log weekly wage earnings, following the methodology of Acemoglu and Autor (2011) and Autor (2019). The series represent composition-adjusted mean log wages for gender-education-experience groups, categorized into four education levels (high school dropout, high school graduate, some college, and college graduate or higher) and five potential experience brackets (0–9, 10–19, 20–29, 30–39, and 40+ years). Log weekly wages are regressed annually by gender on education dummies, a quartic function of experience, and their interactions. The composition-adjusted mean log wage for each of the 40 groups is derived as the predicted log wage at specific experience and education levels. Broader group means are calculated as weighted averages of these adjusted cell means, using fixed weights based on the mean share of total hours worked by each group from 2002 to 2024. All values are adjusted for inflation using the consumer price index (CPI).

Figure 3A. Returns to Education and Occupation by Quantile for Selected Years



Source: LFS 2002–2024.

Table 1A. RIF Regression of Wage Gini

	2002/11	2012/16	2017/24
Less than Elementary	0.079***	0.075***	0.065***
	(0.004)	(0.003)	(0.002)
Elementary	0.037***	0.045***	0.040***
	(0.003)	(0.003)	(0.002)
Some High School	0.048***	0.054***	0.053***
	(0.003)	(0.003)	(0.002)
Some College	-0.026***	-0.029***	-0.007***
	(0.003)	(0.002)	(0.001)
College & above	0.055***	0.058***	0.079***
	(0.004)	(0.003)	(0.002)
Experience < 10 yrs	-0.033***	-0.039***	-0.040***
	(0.003)	(0.002)	(0.001)
Experience 10-20 yrs	-0.025***	-0.022***	-0.017***
	(0.003)	(0.002)	(0.001)
Experience 30-40 yrs	0.007	0.008**	0.019***
	(0.004)	(0.003)	(0.002)
Experience >= 40 yrs	0.013*	0.012**	0.038***
	(0.005)	(0.004)	(0.002)
High skilled	0.088***	0.086***	0.090***
	(0.004)	(0.003)	(0.002)
Middle skilled routine	-0.111***	-0.103***	-0.084***
	(0.003)	(0.002)	(0.001)
Middle skilled nonroutine	-0.047***	-0.043***	-0.013***
	(0.004)	(0.003)	(0.002)
Low-end services	-0.053***	-0.050***	-0.080***
	(0.003)	(0.003)	(0.002)
High-end services	-0.108***	-0.106***	-0.119***
	(0.004)	(0.004)	(0.002)
Manufacturing	-0.132***	-0.140***	-0.152***
	(0.004)	(0.003)	(0.002)
Public Sector	-0.034***	0.051***	0.055***
	(0.004)	(0.003)	(0.002)
Female	0.031***	0.047***	0.035***
	(0.002)	(0.002)	(0.001)
Constant	0.491***	0.473***	0.424***
	(0.004)	(0.003)	(0.002)
Observations	339,012	189,629	540,424
Adjusted R-squared	0.036	0.103	0.095

Source: LFS 2002–24.

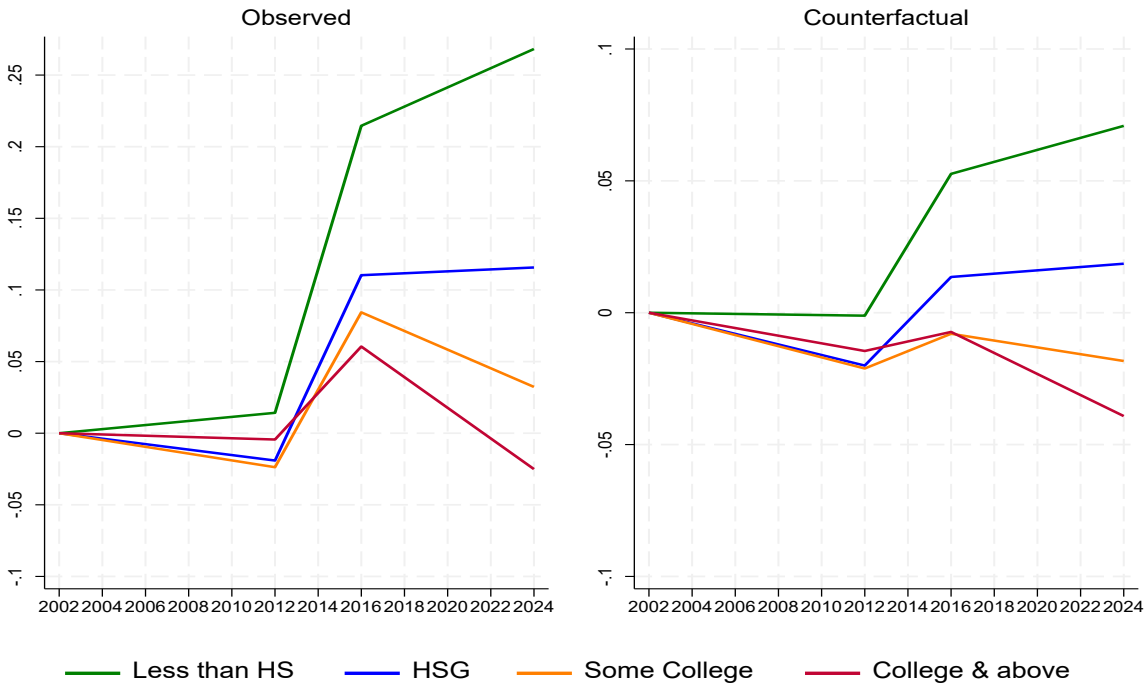
Note: Bootstrapped standard errors, based on 100 repetitions, are reported in parentheses. Significance levels are denoted by: *** $p \leq 0.01$, ** $p \leq 0.05$, * $p \leq 0.1$. The base group consists of individuals who are, not employed in the public sector, hold a high school degree, have 20-30 experience, work in construction, and are in low-skill occupations.

Table 2A. Unconditional Quantile Regression Coefficients on Log Wage

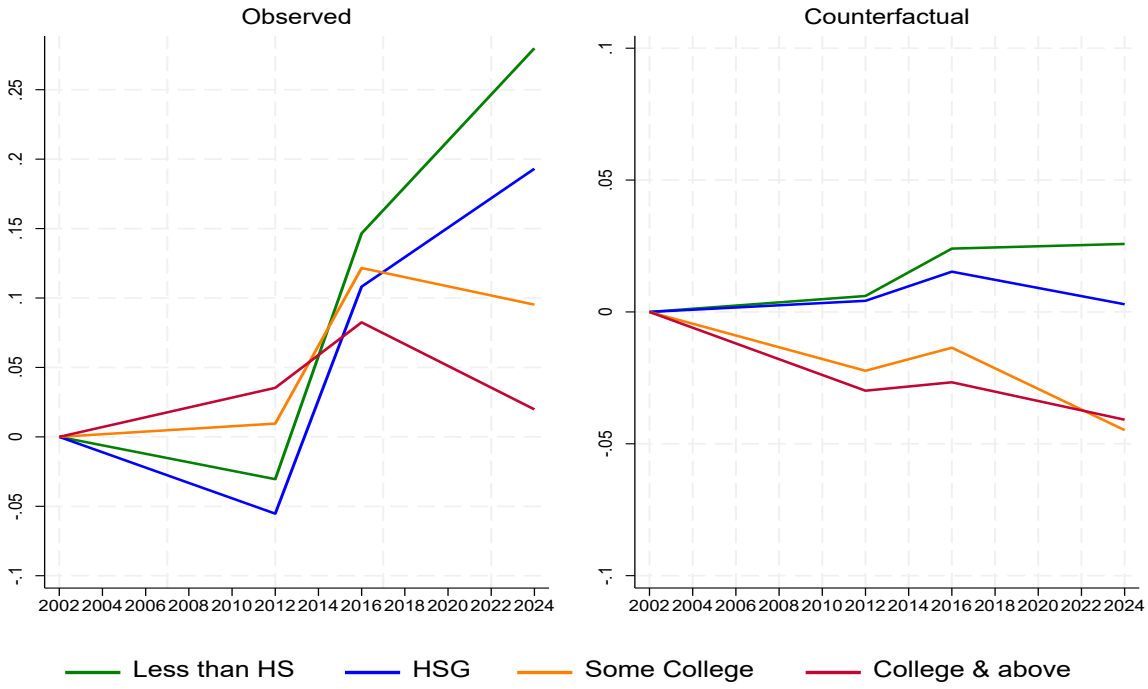
	2002/11			2012/16			2017/24		
	Q10	**Q50**	**Q90**	**Q10**	**Q50**	**Q90**	**Q10**	**Q50**	**Q90**
Below Elem.	-0.472***	-0.366***	-0.065***	-0.544***	-0.299***	-0.023*	-0.460***	-0.185***	-0.020***
	(0.008)	(0.005)	(0.006)	(0.013)	(0.006)	(0.010)	(0.008)	(0.003)	(0.005)
Elementary	-0.220***	-0.282***	-0.069***	-0.285***	-0.237***	-0.029**	-0.291***	-0.139***	-0.020***
	(0.008)	(0.005)	(0.006)	(0.013)	(0.006)	(0.009)	(0.008)	(0.003)	(0.005)
Some HS	-0.267***	-0.245***	-0.031***	-0.350***	-0.221***	-0.013	-0.334***	-0.131***	-0.003
	(0.007)	(0.005)	(0.006)	(0.012)	(0.006)	(0.009)	(0.007)	(0.003)	(0.004)
Some College	0.101***	0.232***	0.097***	0.137***	0.209***	0.086***	0.120***	0.102***	0.100***
	(0.007)	(0.005)	(0.005)	(0.011)	(0.005)	(0.008)	(0.007)	(0.002)	(0.004)
College & above	0.240***	0.525***	0.763***	0.323***	0.478***	0.877***	0.365***	0.297***	0.659***
	(0.008)	(0.005)	(0.006)	(0.013)	(0.006)	(0.009)	(0.007)	(0.002)	(0.004)
Exper < 10 yrs	-0.129***	-0.166***	-0.252***	-0.127***	-0.143***	-0.214***	-0.137***	-0.107***	-0.180***
	(0.006)	(0.004)	(0.005)	(0.010)	(0.005)	(0.007)	(0.006)	(0.002)	(0.004)
Exper 10-20	0.004	-0.018***	-0.104***	0.015	-0.016**	-0.026***	0.017**	-0.021***	-0.019***
	(0.006)	(0.004)	(0.005)	(0.010)	(0.005)	(0.008)	(0.006)	(0.002)	(0.004)
Exper 30-40	-0.017*	0.002	0.036***	-0.011	-0.001	0.003	-0.048***	-0.008**	-0.008
	(0.008)	(0.005)	(0.006)	(0.013)	(0.006)	(0.009)	(0.007)	(0.003)	(0.004)
Exper >= 40	-0.145***	-0.059***	-0.078***	-0.214***	-0.071***	-0.095***	-0.331***	-0.066***	-0.064***
	(0.011)	(0.007)	(0.009)	(0.018)	(0.008)	(0.013)	(0.009)	(0.003)	(0.006)
High skilled	0.258***	0.687***	0.904***	0.348***	0.563***	1.003***	0.352***	0.387***	0.723***
	(0.009)	(0.006)	(0.007)	(0.014)	(0.007)	(0.010)	(0.008)	(0.003)	(0.005)
Middle skilled routine	0.308***	0.484***	0.006	0.406***	0.411***	0.012	0.441***	0.239***	0.010**
	(0.006)	(0.004)	(0.005)	(0.009)	(0.005)	(0.007)	(0.005)	(0.002)	(0.003)
Middle skilled nonroutine	0.176***	0.366***	0.160***	0.230***	0.294***	0.121***	0.215***	0.140***	0.127***
	(0.009)	(0.006)	(0.007)	(0.014)	(0.007)	(0.010)	(0.008)	(0.003)	(0.005)
Low-end services	0.392***	0.172***	0.017**	0.400***	0.139***	-0.043***	0.608***	0.149***	-0.032***
	(0.008)	(0.005)	(0.006)	(0.013)	(0.006)	(0.009)	(0.008)	(0.003)	(0.005)
High-end services	0.577***	0.486***	0.101***	0.632***	0.425***	0.084***	0.892***	0.367***	0.079***
	(0.010)	(0.006)	(0.008)	(0.016)	(0.007)	(0.012)	(0.010)	(0.003)	(0.006)
Manufacturing	0.622***	0.534***	0.026***	0.742***	0.479***	0.001	1.004***	0.348***	0.009
	(0.008)	(0.005)	(0.006)	(0.013)	(0.006)	(0.009)	(0.008)	(0.003)	(0.005)
Public Sector	-0.051***	-0.127***	0.052***	-0.072***	-0.154***	0.543***	-0.130***	-0.131***	0.312***
	(0.008)	(0.005)	(0.006)	(0.013)	(0.006)	(0.010)	(0.007)	(0.003)	(0.005)
Female	-0.302***	-0.217***	-0.103***	-0.394***	-0.203***	-0.040***	-0.428***	-0.141***	-0.069***
	(0.005)	(0.003)	(0.004)	(0.007)	(0.004)	(0.005)	(0.004)	(0.002)	(0.003)
Constant	5.782***	6.817***	7.934***	5.727***	6.896***	7.861***	5.791***	7.146***	7.932***
	(0.008)	(0.005)	(0.007)	(0.013)	(0.006)	(0.010)	(0.009)	(0.003)	(0.005)
Observations	339,013	339,013	339,013	189,629	189,629	189,629	540,424	540,424	540,424
Ad. R-squared	0.127	0.389	0.299	0.128	0.354	0.343	0.139	0.282	0.276

Figure 4A. Observed and Counterfactual Changes in Log Wages by Gender, 2002-2024

A. Men

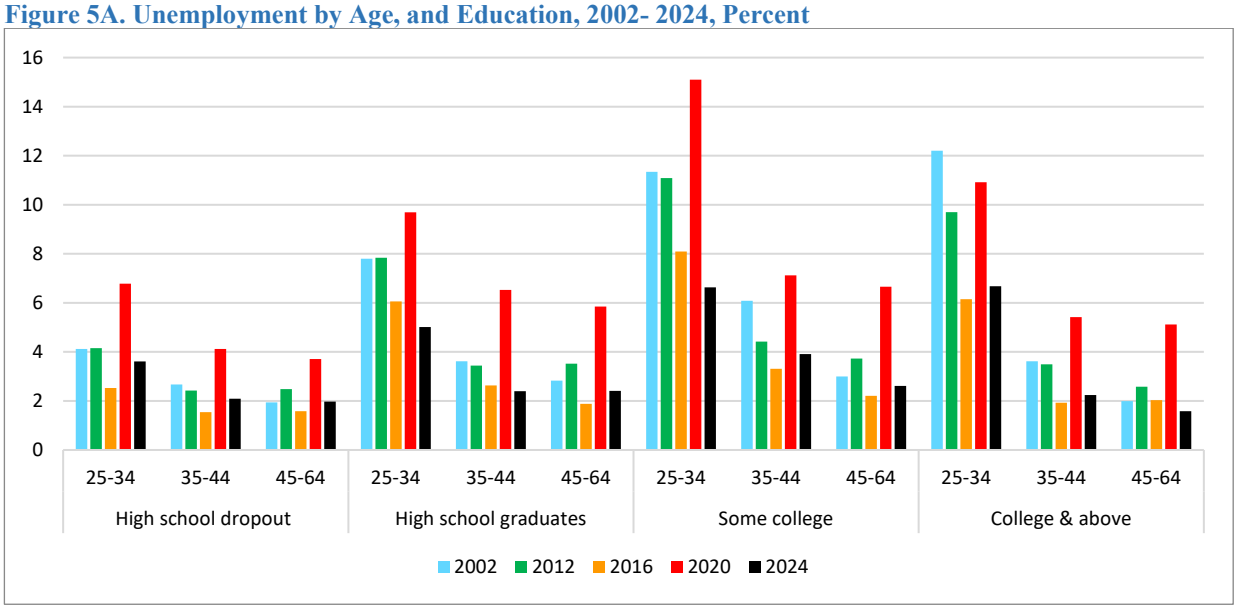


B. Women



Source: LFS 2002–2024.

Figure 5A. Unemployment by Age, and Education, 2002- 2024, Percent



Source: LFS 2002, 2012, 2016, 2020, 2024.

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